

The category of representations of the symmetric group S_t , when t is not a natural integer

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Abstract. Let k be a field of characteristic 0. For $t \in k$ which is not an integer ≥ 0 , we construct a semi-simple tensor category $\text{Rep}(S_t)$ which “interpolates” the categories of linear representations (over k) of the symmetric groups S_n , for n large. When t is a natural integer, $\text{Rep}(S_t)$ can be “specialized” to a tensor category in at least two ways: as the usual category of representations of S_t , or as a non-semisimple tensor category.

The theory reproduces, in a simpler form, known analogous results for the linear or orthogonal groups, which will be recalled.

Introduction

If G is an affine group scheme over a field k , the category $\text{Rep}(G)$ of finite-dimensional linear representations of G over k is an abelian category. It is endowed with the tensor-product functor for representations

$$\otimes : \text{Rep}(G) \times \text{Rep}(G) \longrightarrow \text{Rep}(G).$$

This functor is ACU, i.e. it is endowed with associativity and commutativity constraints, and it has a unit object $\mathbf{1}$. It is rigid: each object X has a dual $X^\vee$, an object endowed with

$$\delta : \mathbf{1} \longrightarrow X \otimes X^\vee, \quad \text{ev} : X^\vee \otimes X \longrightarrow \mathbf{1}$$

such that the composite morphisms

$$X \xrightarrow{\delta \otimes X} X \otimes X^\vee \otimes X \xrightarrow{X \otimes \text{ev}} X \quad X^\vee \xrightarrow{X^\vee \otimes \delta} X^\vee \otimes X \otimes X^\vee \xrightarrow{\text{ev} \otimes X^\vee} X^\vee$$

are identities. Finally,

$$k \xrightarrow{\sim} \text{End}(\mathbf{1}).$$

In other words, in the terminology of (Deligne 1990), 2.1, $\text{Rep}(G)$ is a tensor category over k .

If k is algebraically closed, the tensor category $\text{Rep}(G)$ determines G up to isomorphism (Saavedra 1972). From this point of view, tensor categories generalize affine group schemes. There are examples of tensor categories other than the $\text{Rep}(G)$, but one has no idea how to classify them all, not even those which are semisimple and finitely $\otimes$ -generated (and hence analogous to categories $\text{Rep}(G)$ for G an algebraic group which is an extension of a finite group by a reductive group).

Another example of a tensor category is the category of super-representations of an affine super group scheme G . A more natural variant is to consider G endowed with $\varepsilon : \mu_2 \rightarrow G$ such that the inner action of μ_2 on the affine algebra $\mathcal{O}(G)$ of G defines the modulo 2 grading of $\mathcal{O}(G)$, and the category $\text{Rep}(G, \varepsilon)$ of super-representations V for which the action of μ_2 gives the modulo 2 grading of V .

In characteristic p , S. Gelfand and D. Kazhdan (1992) constructed new semisimple tensor categories as “subquotients” of representation categories. By contrast, over an algebraically closed field k of characteristic 0, the only tensor categories that I know are obtained by “interpolation” of categories $\text{Rep}(G, \varepsilon)$.

Here is a method of interpolation. Start from an infinite family of tensor categories C_i over k , from a non-trivial ultrafilter F on the index set I , and take the ultraproduct C of the C_i . An object of C is therefore a family (X_i) of objects of the C_i , a morphism $(X_i) \rightarrow (Y_i)$ is a germ, with respect to F , of families of morphisms $X_i \rightarrow Y_i$, and the tensor product is given by

$$(X_i) \otimes (Y_i) = (X_i \otimes Y_i).$$

The category C is tensorial over the field k_F , the ultrapower of k . It is not very friendly, since, even if the C_i are categories of representations of linear algebraic groups, the Hom-groups in C will be infinite-dimensional. To do better, one can start from tensor categories C_i over k , each endowed with an object X_i , and consider the full tensor subcategory C' of C generated by $X = (X_i)$. The objects of C' are then the subquotients of sums of

$$T^{r,s}(X) := X^{\otimes r} \otimes (X^\vee)^{\otimes s}.$$

Assume that

(*) for all r and s , the object $T^{r,s}(X_i)$ of C_i has finite length, bounded independently of i .

Under this hypothesis the objects of C' have finite length, and the Hom-groups are therefore finite-dimensional over k_F (consider $\text{Hom}(X, Y)$ and apply 3.2.4).

The definition by ultraproduct is somewhat theological. In good cases, one can associate with each i a parameter λ_i in k describing (C_i, X_i) , and C' can be constructed purely in terms of $\lambda = (\lambda_i) \in k_F$. More precisely, for k_1 an extension of k and $\lambda_1 \in k_1$ transcendental over k , one can construct a tensor category $C(\lambda_i)$ over k_1 , which gives C' when one takes $k_1 = k_F$ and $\lambda_1 = (\lambda_i)$. The role of ultraproducts is then only to have drawn attention to condition (*).

The case where λ_1 is transcendental is the easy case. More interesting, and more difficult, is to take λ_1 arbitrary in an extension k_1 of k , or at least to understand which values of λ_1 are singular, and how.

From now on k is assumed to have characteristic 0. The case treated in this article is:

I : the set of integers ≥ 0 ;
 C_n : the category of representations of S_n ;
 X_n : the permutation representation on k^n ;
parameter: n ;
singular values: the integers ≥ 0 .

The following cases are essentially known and will be recalled:

I = the set of integers ≥ 0 ;
 C_n = $\text{Rep}(\text{GL}(n))$ (resp. $\text{Rep}(O(n))$);
 X_n = the evident representation;
parameter: n ;
singular values: $\mathbb{Z}$.

When C_i is the category of representations of a finite group G_i , and X_i is the permutation representation defined by an action of G_i on a finite set B_i , condition (*) is equivalent to

(*)' for every r , the number of orbits of G_i in B_i^r is bounded independently of i .

Here are examples in which condition (*)' is satisfied. In each of them one fixes a finite field $\mathbb{F}_q$, and I is the set of integers ≥ 0 :

- (A) $G_n = \text{GL}(n, \mathbb{F}_q)$, $B_n = \mathbb{F}_q^n$;
- (B) $G_n = O(n, \mathbb{F}_q)$, $B_n = \mathbb{F}_q^n$;
- (C) $G_n = \text{Sp}(2n, \mathbb{F}_q)$, $B_n = \mathbb{F}_q^{2n}$.

In case (A), one can take the parameter $\lambda_n := q^n$, and I conjecture that the singular values of the parameter are the q^n , for $n \geq 0$.

I thank the referee for a careful reading, which led me to correct several obscure points.

1. Terminology and notation

1.1. We shall use the following notation.

k : a field of characteristic 0, fixed throughout. For I a finite set:

$$\begin{aligned} S_I &: \text{ the symmetric group of permutations of } I; \\ k^I &: \text{ the } k\text{-vector space freely generated by } I; \\ (e_i)_{i \in I} &: \text{ its evident basis.} \end{aligned}$$

If $I = \{1, \dots, n\}$, we write S_n (resp. k^n) instead of S_I (resp. k^I).

1.2. Let $(\cdot, \cdot)$ be the symmetric bilinear form on k^I defined by $(e_i, e_j) = \delta_{ij}$. It identifies k^I with its dual. Viewing k^I as the space of functions from I to k , via

$$u \mapsto \sum u(i)e_i,$$

one defines an algebra structure on k^I by

$$(u \cdot v)(i) = u(i)v(i).$$

For this algebra structure,

$$(u, v) = \text{Tr}(uv).$$

1.3. The functor $I \mapsto k^I$, being a left adjoint, preserves inductive limits, and in particular carries disjoint sums to direct sums. It carries finite products to tensor products, by

$$\bigotimes_{j \in J} k^{I_j} \xrightarrow{\sim} k^{\prod I_j}, \quad \bigotimes e_{a(j)} \mapsto e_{(a(j))_{j \in J}}.$$

1.4. If a group G acts on I , it acts on k^I by transport of structure:

$$ge_i = e_{gi}.$$

The linear representation k^I of G is called the permutation representation defined by the G -set I . The form $(\cdot, \cdot)$ and the algebra structure of 1.2 are G -invariant. The functor $I \mapsto k^I$ still carries disjoint sums (resp. finite products) of G -sets to direct sums (resp. tensor products) of representations.

1.5. Let $(U_j)_{j \in J}$ be a finite family of finite sets. A gluing of the U_j is a finite set A endowed with a family of injections

$$f_j : U_j \hookrightarrow A$$

such that A is the union of the $f_j(U_j)$.

The U_j being fixed, a gluing datum on the U_j is an isomorphism class of gluings. If r is a gluing datum, and $(f_j : U_j \hookrightarrow A)$ is a representative of r , then A is determined, up to unique isomorphism, by r . This makes harmless the abuse of language, which we shall allow ourselves,

of saying “gluing” for “gluing datum”. For example, saying “the sum over gluings $(f_j : U_j \hookrightarrow A)$ of the U_j of $\varphi(A)$ ” means “the sum over gluing data r on the U_j of $\varphi(A_r)$, for $(f_j : U_j \hookrightarrow A_r)$ a representative of r ”.

A gluing datum on the U_j identifies with an equivalence relation R on the disjoint sum of the U_j which induces the discrete equivalence $x = y$ on each U_j : to R , attach the maps

$$U_j \hookrightarrow (\coprod U_j)/R.$$

For $J = \{1, 2\}$, it also identifies with the data of subsets $U'_j \subset U_j$ and of a bijection

$$\varphi : U'_1 \longrightarrow U'_2 :$$

attach to it the maps $U_j \hookrightarrow U_1 \coprod_{\varphi} U_2$.

For $K \subset J$ and $(f_j : U_j \hookrightarrow A)$ a gluing of the U_j , the induced gluing of the U_j , $j \in K$, is the family

$$f_j : U_j \hookrightarrow \bigcup_{j \in K} f_j(U_j).$$

1.6. Let A be a commutative ring. A category $\mathcal{A}$ is said to be A -linear if the Hom-groups are endowed with A -module structures for which composition is bilinear. For B a commutative A -algebra, $\mathcal{A} \otimes_A B$ is then the category deduced from $\mathcal{A}$ by extension of scalars from A to B : the objects are the same, and

$$\mathrm{Hom}(X, Y) = \mathrm{Hom}_A(X, Y) \otimes_A B.$$

1.7. Let $\mathcal{A}$ be A -linear. The additive envelope $\mathcal{A}^{\mathrm{add}}$ of $\mathcal{A}$ is the additive A -linear category of finite formal sums of objects of $\mathcal{A}$:

$$\begin{array}{ll} \text{objects:} & \text{finite families } (A_j)_{j \in J} \text{ of objects of } \mathcal{A}; \\ \text{morphisms } (A_j)_{j \in J} \rightarrow (B_k)_{k \in K} : & \text{a matrix } (f_k^j) \text{ of morphisms } A_j \rightarrow B_k; \\ \text{composition:} & \text{matrix multiplication.} \end{array}$$

1.8. A pseudo-abelian category (SGA4 I 8.7.8 says “additive and Karoubian”) is an additive category in which every idempotent endomorphism is the projection onto a direct factor. The pseudo-abelian envelope $\mathcal{A}^{\mathrm{psab}}$ of an A -linear category $\mathcal{A}$ is the pseudo-abelian A -linear category of formal direct factors (images of idempotent endomorphisms) of finite formal sums of objects of $\mathcal{A}$. See SGA4 loc. cit.

1.9. Let $\mathrm{Hom}_A(\mathcal{A}_1, \mathcal{A}_2)$ denote the category of A -linear functors from $\mathcal{A}_1$ to $\mathcal{A}_2$. The constructions 1.7 and 1.8 have a universal property: if $\mathcal{B}$ is A -linear and additive (resp. pseudo-abelian), the restriction functor

$$\mathrm{Hom}_A(\mathcal{A}^{\mathrm{add}}, \mathcal{B}) \longrightarrow \mathrm{Hom}_A(\mathcal{A}, \mathcal{B})$$

(resp.

$$\mathrm{Hom}_A(\mathcal{A}^{\mathrm{psab}}, \mathcal{B}) \longrightarrow \mathrm{Hom}_A(\mathcal{A}, \mathcal{B}))$$

is an equivalence. It is “the same thing” to give an A -linear functor from $\mathcal{A}$ to $\mathcal{B}$, or from $\mathcal{A}^{\mathrm{add}}$ (resp. $\mathcal{A}^{\mathrm{psab}}$) to $\mathcal{B}$.

An A -bilinear functor

$$\mathcal{A} \times \mathcal{A} \longrightarrow \mathcal{A}^{\mathrm{add}}$$

therefore extends to an A -bilinear functor

$$\mathcal{A}^{\text{add}} \times \mathcal{A}^{\text{add}} \longrightarrow \mathcal{A}^{\text{add}},$$

and this extension is unique up to unique isomorphism. The same holds with “add” replaced by “ps ab”.

If the A -linear category $\mathcal{A}$ is endowed with an A -bilinear ACU tensor product (associative, commutative, with unit), then $\mathcal{A}^{\text{add}}$ and $\mathcal{A}^{\text{ps ab}}$ inherit it. If the tensor product of $\mathcal{A}$ is rigid, then that of $\mathcal{A}^{\text{add}}$ (resp. $\mathcal{A}^{\text{ps ab}}$) is rigid as well.

1.10. A partition λ is a decreasing sequence of integers $(\lambda_a)_{a \geq 1}$, zero for a sufficiently large. Put

$$|\lambda| = \sum_a \lambda_a,$$

and say that λ is a partition of ℓ if $|\lambda| = \ell$. A sequence $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r$ is viewed as a partition by prolonging it by zeros. The transposed partition λ^* of a partition λ is characterized by the equivalence between $\lambda_i \geq j$ and $\lambda_j^* \geq i$. If λ is a partition of ℓ , we shall also write λ for the following objects attached to λ .

- (A) The corresponding isomorphism class of irreducible representations of S_ℓ ; its class in the Grothendieck group $R(S_\ell)$ of representations of S_ℓ , and its character. The trivial representation of S_ℓ corresponds to the coarse partition ℓ of ℓ .
- (B) For N large enough that $\lambda_{N+1} = 0$: the dominant weight $(\lambda_1, \dots, \lambda_N)$ of $\text{GL}(N)$.

For each ℓ and each partition λ of ℓ , assume chosen an irreducible representation of class λ of S_ℓ . The category of sets U with ℓ elements and bijections between them being equivalent to its full subcategory reduced to the object $\{1, \dots, \ell\}$, it is the same to give such a representation V_λ , or a functor $U \mapsto V_\lambda[U]$. In particular, the choice of V_λ defines, for every set U with ℓ elements, a representation of S_U , which we shall again denote by λ . It is deduced from V_λ by twisting by the S_ℓ -torsor of bijections from $\{1, \dots, \ell\}$ to U .

In the preceding discussion, the base field is $\mathbb{Q}$. One could even take the base ring to be $\mathbb{Z}(1/\ell!)$. The case of other base fields or rings is obtained by extension of scalars. If X is a representation of S_U , put

$$X_\lambda := \text{Hom}(\lambda, X). \tag{1.10.1}$$

It is known that

$$\mathbb{Q}[S_U] \xrightarrow{\sim} \bigoplus_{|\lambda|=\ell} \text{End}(\lambda), \tag{1.10.2}$$

and it follows that the natural S_U -equivariant map

$$\bigoplus_{|\lambda|=\ell} X_\lambda \otimes \lambda \xrightarrow{\sim} X \tag{1.10.3}$$

is an isomorphism. This construction still makes sense for X in a pseudo-abelian $\mathbb{Z}(1/\ell!)$ -linear category.

2. The category $\text{Rep}(S_T)$

2.1. Fix a finite set I . For U a finite set, write $[U]_I$ for the permutation representation of S_I on the set $\text{Inj}(U, I)$ of injections from U into I . When U (resp. I) is the finite set $\{1, 2, \dots, n\}$, U (resp. I) is replaced by n in the notation.

For an injection $\varphi : U \hookrightarrow V$, composition with φ ,

$$\text{Inj}(V, I) \longrightarrow \text{Inj}(U, I),$$

induces

$$\text{res}_\varphi : [V]_I \longrightarrow [U]_I$$

whose transpose (1.2) is

$$\text{res}_\varphi^* : [U]_I \longrightarrow [V]_I.$$

One has

$$\text{res}_\varphi(e_g) = e_{g\varphi}, \quad \text{res}_\varphi^*(e_f) = \sum_{g\varphi=f} e_g.$$

2.2. Let C be a gluing of the finite sets U and V , and let D be the fibre product of U and V over C :

$$\begin{array}{ccc} D & \xrightarrow{a'} & U \\ b' \downarrow & & \downarrow a \\ V & \xrightarrow{b} & C. \end{array} \quad (2.2.1)$$

If $U' := a'(D)$, $V' := b'(D)$, and if φ is the bijection $b'a'^{-1} : U' \xrightarrow{\sim} V'$, then

$$U \coprod_{\varphi} V \xrightarrow{\sim} C.$$

Put

$$\{C\}_I := \text{res}_{b'}^* \text{res}_{a'} \quad (\text{resp. } (C)_I := \text{res}_b \text{res}_a^*) : [U]_I \longrightarrow [V]_I.$$

These morphisms will also be denoted $\{\varphi\}_I$ and $(\varphi)_I$, and, when this creates no ambiguity, the index I will be omitted.

Define an ordering on gluing data of U and V by saying that A is coarser than C ($A \leq C$) if, with D and φ as above, the diagram

$$\begin{array}{ccc} D & \longrightarrow & U \\ \downarrow & & \downarrow \\ V & \longrightarrow & A \end{array}$$

is commutative; equivalently, A is defined by a bijection

$$\psi : U'' \xrightarrow{\sim} V'',$$

with $U' \subset U'' \subset U$ and $V' \subset V'' \subset V$, extending φ .

For $f \in \text{Inj}(U, I)$, the image of e_f under $\{C\}$ (resp. under (C)) is the sum of the e_g , for $g \in \text{Inj}(V, I)$, such that f and g glue to a map (resp. an injective map) from C to I . This description shows that

$$\{C\} = \sum_{A \leq C} (A). \quad (2.2.2)$$

If $A \leq C$ is defined by $\psi : U'' \xrightarrow{\sim} V''$, the ordered set of gluings A_1 between A and C is isomorphic to the set, ordered by $\supset$, of subsets of $U'' - U'$: to A_1 , defined by $\psi_1 : U_1'' \rightarrow V_1''$, where $U' \subset U_1'' \subset U''$ and ψ_1 is the restriction of ψ , attach $U_1'' - U'$. The Möbius function for the ordered set of gluing data is therefore

$$\mu(A, C) = (-1)^{|U''| - |U'|} = (-1)^{|C| - |A|},$$

and (2.2.2) is inverted as

$$(C) = \sum_{A \leq C} (-1)^{|C|-|A|} \{A\}. \quad (2.2.3)$$

2.3. Special cases:

- (i) If in (2.2.1) a' (resp. b') is bijective, so that the gluing C is defined by $\varphi : U \hookrightarrow V$ (resp. $\varphi : V \hookrightarrow U$), then

$$(C) = \{C\} = \text{res}_\varphi^* \quad (\text{resp. } (C) = \{C\} = \text{res}_\varphi).$$

- (ii) If the gluing is defined by a bijection φ from U to V , then (φ) is the functoriality of $U \mapsto [U]_I$ for bijections, as defined by transport of structure.

Construction (2.4). *Let $(U_j)_{j \in J}$ be a finite family of finite sets. The tensor product of the $[U_j]_I$ is naturally isomorphic to the direct sum, over gluings $(u_j : U_j \hookrightarrow C)$, of the $[C]_I$.*

Proof. The tensor product of the $[U]_I$ is the permutation representation defined by the S_I -set product of the $\text{Inj}(U_j, I)$. Each system of injections $u_j : U_j \hookrightarrow I$ defines the gluing

$$u_j : U_j \hookrightarrow \bigcup u_j(U_j).$$

The corresponding gluing datum depends only on the S_I -orbit of $(u_j)_{j \in J}$, and this makes $\coprod \text{Inj}(U_j, I)$ the disjoint sum of the $\text{Inj}(C, I)$, for $(a_j : U_j \hookrightarrow C)$ a gluing datum. The conclusion follows from 1.3. $\square$

Examples (2.5). (i) *The representation $[\emptyset]_I$ is the unit representation: k with the trivial action of S_I . The corresponding morphism*

$$[U]_I \otimes [\emptyset]_I \xrightarrow{\sim} [U]_I$$

is defined by the evident gluing of U and $\emptyset$ into U .

- (ii) *The algebra structure*

$$[U]_I \otimes [U]_I \longrightarrow [U]_I$$

of $[U]_I$ defined in 1.2 is the projection onto the direct summand corresponding to the evident gluing of U and U into U .

- (iii) *The self-duality 1.2 of $[U]_I$ is*

$$[U]_I \otimes [U]_I \xrightarrow{(ii)} [U]_I \xrightarrow{\text{res}_{\emptyset \rightarrow U}} [\emptyset],$$

and the corresponding morphism $\delta : \mathbf{1} \rightarrow [U]_I \otimes [U]_I$ is

$$[\emptyset] \xrightarrow{\text{res}_{\emptyset \rightarrow U}^*} [U]_I \longrightarrow [U]_I \otimes [U]_I,$$

where the second arrow is the inclusion of the direct summand $[U]_I$.

2.6. Let $((U_{j,k})_{j \in J_k})_{k \in K}$ be a family of families. It defines the family of the $U_{j,k}$, indexed by $\coprod_k J_k$. There is a natural bijection between:

- (A) gluing data C on the $U_{j,k}$;
- (B) systems formed by:

- (a) for each k , a gluing datum C_k on the $U_{j,k}$, $j \in J_k$;
- (b) a gluing datum D on the C_k , $k \in K$.

To C , one attaches the induced gluings, and their gluing into C .

We leave to the reader the verification that the isomorphism

$$\bigotimes_{k \in K} \bigotimes_{j \in J_k} [U_{j,k}]_I = \bigotimes [U_{j,k}]_I$$

becomes, via the isomorphisms 2.4, the isomorphism of

$$\bigoplus_{(C_k)} \bigotimes [C_k]_I = \bigoplus_{(D, C_k)} [D]_I$$

with the sum of the $[C]_I$, defined by the bijections $(A) \leftrightarrow (B)$.

2.7. This describes the associativity and commutativity constraints and the unit of the tensor product of representations $[U]_I$ in terms of the description 2.4 of the tensor product. For associativity

$$([U]_I \otimes [V]_I) \otimes [W]_I \longrightarrow [U]_I \otimes ([V]_I \otimes [W]_I),$$

one identifies both sides with the sum of the $[C]_I$, for C a gluing of U, V, W . For commutativity

$$[U]_I \otimes [V]_I \longrightarrow [V]_I \otimes [U]_I,$$

use the fact that a gluing of U and V identifies with a gluing of V and U . The unit is $[\emptyset]_I$.

Proposition (2.8). *Let $(U_j)_{j \in J}$ and $(V_j)_{j \in J}$ be two families indexed by J , and, for each j , let $U_j, V_j \hookrightarrow C_j$ be a gluing of U_j and V_j . Using 2.3, identify the tensor product of the $[U_j]$ (resp. $[V_j]_I$) with the sum of the $[A]_I$ (resp. $[B]_I$), for A (resp. B) a gluing of the U_j (resp. V_j). Then the matrix coefficient*

$$[A]_I \longrightarrow \bigotimes [U_j]_I \longrightarrow \bigotimes [V_j]_I \longrightarrow [B]_I$$

of

$$\bigotimes (C_j) : \bigotimes [U_j]_I \longrightarrow \bigotimes [V_j]_I$$

is the sum of the

$$(C) : [A]_I \longrightarrow [B]_I,$$

where C ranges over gluing data on A and B which, viewed by 2.6 as gluing data on the U_j and V_j , induce for each j the gluing datum C_j of U_j and V_j .

The verification, immediate from the description 2.2 of the (C_j) , is left to the reader.

Proposition (2.9). (i) *The morphisms $(C) : [U]_I \rightarrow [V]_I$, where $U, V \hookrightarrow C$ is a gluing of U and V such that $|C| \leq |I|$, form a basis of $\text{Hom}_{S_I}([U]_I, [V]_I)$.*

(ii) *The condition $|C| \leq |I|$ is satisfied by every gluing $U, V \hookrightarrow C$ if $|U| + |V| \leq |I|$.*

Proof. By self-duality of the permutation representation $[U]_I$, the group $\text{Hom}_{S_I}([U]_I, [V]_I)$ identifies with the space of S_I -invariants in $[U]_I \otimes [V]_I$, which is the permutation representation defined by $\text{Inj}(U, I) \times \text{Inj}(V, I)$. The S_I -invariants have a basis indexed by the S_I -orbits: to an orbit K corresponds the sum of the e_k , for $k \in K$. Injections $u : U \hookrightarrow I$ and $v : V \hookrightarrow I$ define the gluing datum

$$U, V \hookrightarrow u(U) \cup v(V).$$

Denote it $C_{u,v}$. It depends only on the orbit of (u, v) , and detects it. We leave to the reader the verification that the morphism from $[U]_I$ to $[V]_I$ defined by the orbit of (u, v) is $(C_{u,v})$. The gluing data C so obtained are those for which $|C| \leq |I|$. For the others, $[C] = 0$. This proves (i), and (ii) is trivial. $\square$

Proposition (2.10). *Let U, V, W be three finite sets. Let $U, V \hookrightarrow C$ be a gluing of U and V , and let $V, W \hookrightarrow D$ be a gluing of V and W . Then the composite morphism*

$$(D)(C) : [U]_I \longrightarrow [W]_I$$

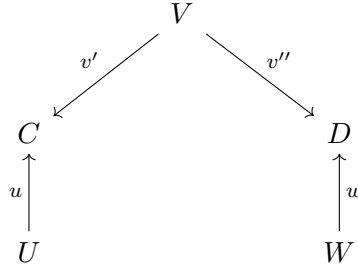
is the sum, over gluings $a, b, c : U, V, W \hookrightarrow A$ which induce C and D ,

$$(D)(C) = \sum_A P_A(|I|)(A'), \quad (2.10.1)$$

where A' is the gluing of U and W induced by A , and where $P_A(T)$ is the polynomial with integer coefficients

$$P_A(T) := (T - |A'|)(T - |A'| - 1) \cdots (T - |A| + 1). \quad (2.10.2)$$

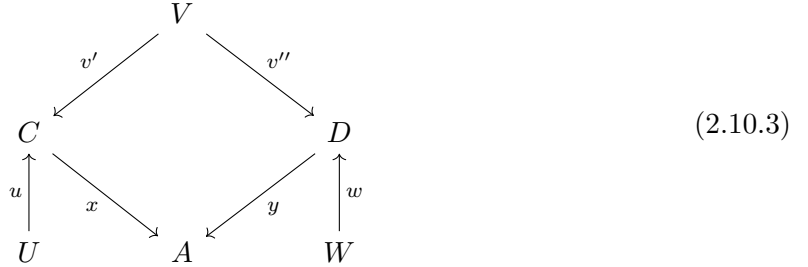
Proof. Write



for the gluings. By definition (2.2),

$$(D)(C) = \text{res}_w \text{res}_{v''}^* \text{res}_{v'} \text{res}_u^*.$$

The composite $\text{res}_{v''}^* \text{res}_{v'}$ is computed by (2.2.2): it is the sum, over gluings $x, y : C, D \hookrightarrow A$ of C and D making



commutative, of the $\text{res}_y \text{res}_x^*$.

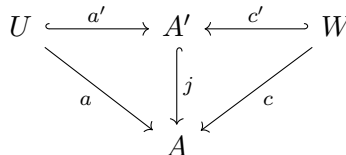
The map which sends $x, y : C, D \hookrightarrow A$ to

$$xu, \quad xv' = yv'', \quad yw : U, V, W \hookrightarrow A$$

is bijective from the gluing data under consideration to the gluing data of U, V, W inducing the given data on U, V and V, W . For each diagram (2.10.3) one has

$$\text{res}_w \text{res}_y \text{res}_x^* \text{res}_u^* = \text{res}_{yw} \text{res}_{xu}^*,$$

and the composite $(D)(C)$ is therefore the sum, over $a, b, c : U, V, W \hookrightarrow A$ inducing C and D , of the $\text{res}_c \text{res}_a^*$. Let $A' := a(U) \cup c(W)$:



Then

$$\text{res}_c \text{res}_a^* = \text{res}_{c'} \text{res}_j \text{res}_j^* \text{res}_{a'}^*,$$

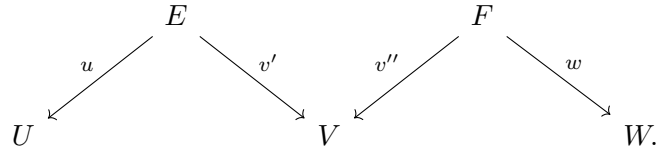
and $\text{res}_j \text{res}_j^*$ is multiplication by the number $P_A(|I|)$ of ways to extend an injection of A' into I to an injection of A into I . Hence

$$(D)(C) = \sum_A \text{res}_c \text{res}_a^* = \sum_A P_A(|I|) \text{res}_{c'} \text{res}_{a'}^* = \sum_A P_A(|I|)(A').$$

□

Remark 2.11. An analogous calculation, with (2.2.2) replaced by (2.2.3), expresses the composition of morphisms $\{C\}$. Assume that

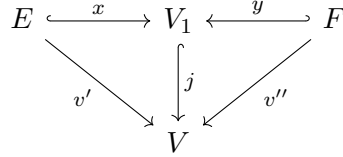
$$C = U \coprod_E V, \quad D = V \coprod_F W :$$



By definition,

$$\{D\}\{C\} = \text{res}_w^* \text{res}_{v''} \text{res}_{v'}^* \text{res}_u.$$

Put $V_1 := v'(E) \cup v''(F)$:



Then

$$\text{res}_{v''} \text{res}_{v'}^* = \text{res}_y \text{res}_j \text{res}_j^* \text{res}_x^* = P_{C,D}(|I|) \text{res}_y \text{res}_x^*,$$

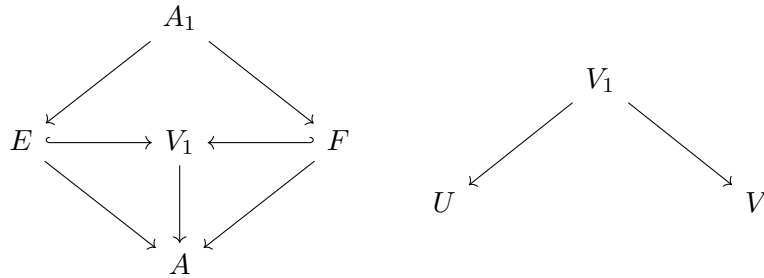
for $P_{C,D}$ the polynomial with integer coefficients

$$P_{C,D}(T) := (T - |V_1|) \cdots (T - |V| + 1). \quad (2.11.1)$$

The composite $\text{res}_y \text{res}_x^*$ is

$$(V_1) : [E]_I \longrightarrow [F]_I.$$

By (2.2.3), it is an alternating sum of morphisms $\{A\}$, for A a gluing of E and F coarser than V_1 . If A_1 is the fibre product $E \times_A F$, i.e. for $A = E \coprod_{A_1} F$,



A_1 defines a gluing A' of U and V , and $\{D\}\{C\}$ is the sum over A

$$\{D\}\{C\} = P_{C,D}(|I|) \sum (-1)^{|V_1| - |A|} \{A'\}. \quad (2.11.2)$$